

Modeling "Iki" as a Coherent Control Signal

Optimization of Control Signals via Integration of Prosocial Homeostasis and Minimum Jerk Trajectories

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Abstract

"Iki," an aesthetic concept from Japan's Edo period, refers to a refined state of mind and body, but its generative mechanism and physical properties have historically been discussed only qualitatively. In this study, we redefine "Iki" not merely as a subjective aesthetic sense, but as an "Effective Control Signal (S_{eff})" that enables an individual to achieve physical synchronization with the environment and others. We model its generation process mathematically.

Based on evolutionary biological Prosocial Homeostasis, we focus on the mechanism by which the brain's reward system motivates physical System Optimization (D). In our model, "Iki" is defined as a state where the minimization of Neural Entropy (σ) (sedation of the Default Mode Network) and the maximization of Minimum Jerk (D) (smoothness of motion) are achieved simultaneously. We propose, through a control engineering approach, that this highly efficient signal state serves as the driving force for attractor formation (state determination) in the physical world.

1 Introduction

1.1 Background: The Divergence of Subjectivity and Physics

In the dualistic paradigm since Descartes, individual "Will" and "Aesthetics" have been treated as events independent of objective physical laws. However, in the physical performance of skilled artisans, martial artists, or artists, a highly subjective mental state (e.g., "Mushin" or "Flow") is inseparably linked to physical high efficiency (e.g., motion without waste). This phenomenon suggests that mental variables have a direct influence on physical control parameters.

1.2 Problem Statement

Conventional motor control models and behavioral science models struggle to quantitatively handle mental variables such as "hesitation" or "benevolence" (consideration for others). Consequently, they fail to provide a unified explanation for the generative mechanism of why skilled movements appear "beautiful" (i.e., physically efficient).

1.3 Objective of this Study

This paper mathematically models the traditional Japanese concept of "Iki" as a process of noise reduction and signal optimization. Specifically, by introducing variables for Default Mode Network (DMN) activity from neuroscience and Jerk from kinematics, we argue that spiritual cultivation and physical refinement share the mathematically identical objective function: maximization of the Signal-to-Noise Ratio (SNR).

2 Theoretical Framework

2.1 Prosocial Homeostasis

We define the survival strategy evolved by life in interdependence with the global environment as "Prosocial Homeostasis." Individuals perceive minimizing friction and achieving synchronization with others and the physical environment as "pleasure" (reward) via the brain's reward system (oxytocin, dopamine). This biological basis provides humans with the motivation to execute "aligning with others" not merely as a moral duty, but as a pleasure principle to maximize energy efficiency.

2.2 Minimum Jerk Model

According to the motor control theory proposed by Flash & Hogan (1985), skilled human movements converge to a trajectory that minimizes the sum of the squared jerk (the rate of change of acceleration) from start to finish [1]. Jerk J is expressed by the following equation:

$$J = \int_0^T \left| \frac{d^3 \mathbf{r}}{dt^3} \right|^2 dt \quad (1)$$

In this study, we adopt the reciprocal of this jerk (smoothness) as an index of the individual's "System Optimization Ability" (D) to physically adapt to the environment.

2.3 Neural Entropy and DMN

The Default Mode Network (DMN), identified by Raichle et al. (2001), is a brain network active during rest and is involved in self-referential thinking (reflection, anxiety, ego-consciousness) [2]. It is known that DMN activity decreases during flow states or meditation. In this study, we define excessive DMN activity as "Internal Noise" or "Neural Entropy" that hinders signal transmission to the outside.

3 Mathematical Model: The Iki Generation Process

We define "Iki" as an output state where input energy has been filtered and purified. The generation process is formulated below.

3.1 Variable Definitions

This model defines three primary variables:

1. Drive Potential (E_{in}):

- Definition: Total input energy based on the individual's survival instincts and desires. The initial impulse of "wanting to do."
- Physical Indicator: Arousal of the autonomic nervous system. Measurable via Electrodermal Activity (EDA) or pupil diameter.

2. System Optimization Coefficient (D):

- Definition: The degree of fitness of the input to the physical characteristics of the target (environment/others). The physical entity of what is generally called "benevolence" or "consideration."

- Physical Indicator: Smoothness of motion. Quantified as the convergence to the Minimum Jerk trajectory ($D \propto 1/J$).

3. Neural Entropy (σ):

- Definition: Internal interference hindering signal generation. Noise caused by self-preservation or hesitation.
- Physical Indicator: DMN activity level. Takes a normalized range $0 \leq \sigma \leq 1$.

3.2 Signal Purification Equation

The Effective Control Signal S_{eff} (i.e., "Iki") is the Drive Potential E_{in} multiplied by the optimization function η .

$$S_{\text{eff}} = \eta(\sigma, D) \cdot E_{\text{in}} \quad (2)$$

Here, the Signal Purification Efficiency η is expressed as the product of the complement of Neural Entropy and the System Optimization Coefficient.

$$\eta(\sigma, D) = (1 - \sigma) \cdot D \quad (3)$$

3.3 State Classification

Based on this equation, human performance states are classified as follows:

- Yabo (Unrefined) State [$\sigma \rightarrow 1, D \rightarrow \text{low}$]: Even with high Drive Potential E_{in} , due to high Neural Entropy σ , $\eta \approx 0$. Energy does not become an effective signal but dissipates as heat or destructive impact (high Jerk).
- Iki (Refined) State [$\sigma \rightarrow 0, D \rightarrow \text{high}$]: Neural Entropy is eliminated, and optimization to the system is performed. As $\eta \rightarrow \text{max}$, input energy E_{in} is converted into a coherent output S_{eff} without loss.

3.4 Simulation Verification

To verify the validity of this model, a crowd simulation using 150 autonomous agents was conducted. The results are shown below.

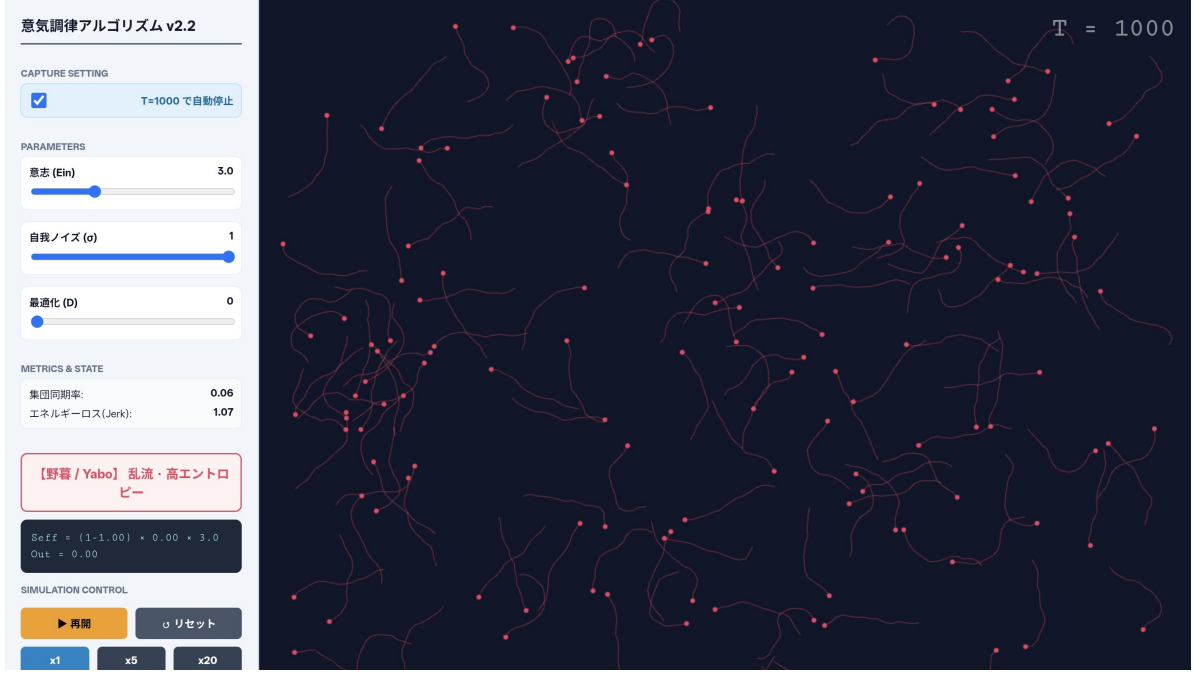


Figure 1: Yabo State ($\sigma = 1.0, D = 0.0, T = 1000$)

Due to high neural entropy, individuals diffuse chaotically. The group synchronization rate is remarkably low at 0.06, and energy loss (Jerk) reaches 1.07, indicating a turbulent state.

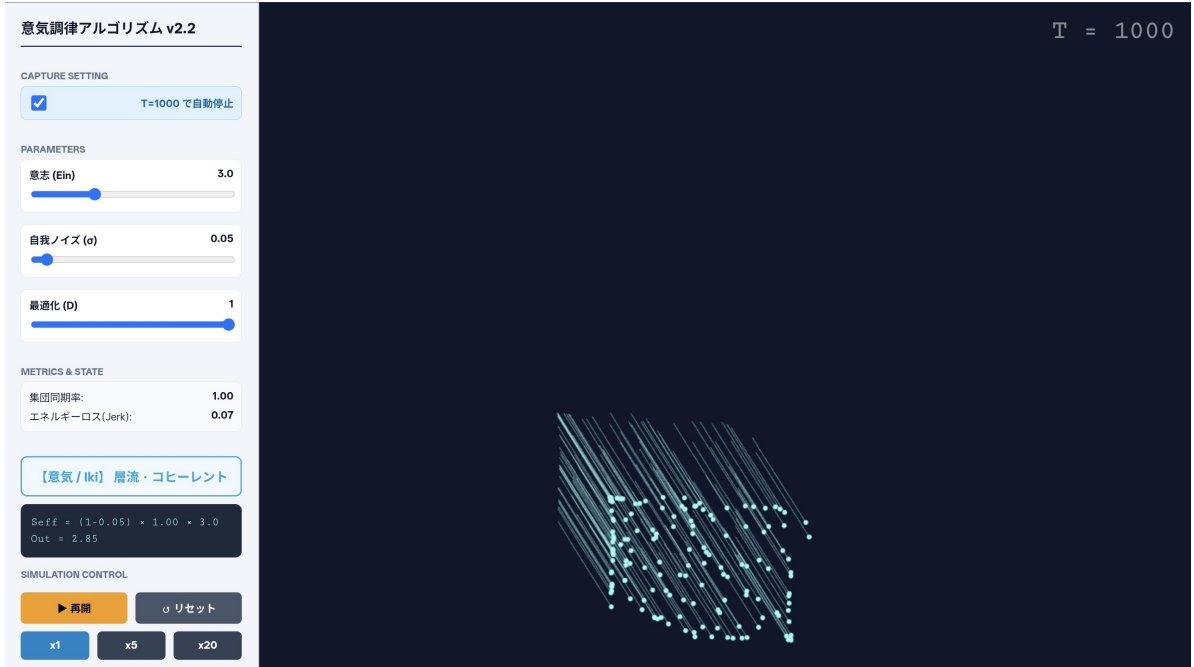


Figure 2: Iki State ($\sigma = 0.05, D = 1.0, T = 1000$)

By maximizing the optimization coefficient, autonomous synchronization occurs. Even at the wall boundary, group cohesion is maintained, and the synchronization rate dramatically improved to 1.00, with Jerk reduced to 0.07.

4 Proposed Validation Methods

We propose the following experimental systems to verify the validity of this model.

4.1 Measurement of "Dynamic Serenity"

We plot the biosignals of skilled experts (e.g., tea masters, veteran drivers) and novices during task execution on the following two axes.

- X-axis (Heat): Amplitude of Electrodermal Activity (EDA).
- Y-axis (Serenity): Reciprocal of Jerk measured by accelerometers.

[Hypothesis] Novices distribute in the "High EDA / High Jerk (Panic)" or "Low EDA / Low Jerk (Relaxed)" regions. In contrast, skilled experts distribute in a unique region of "High EDA / Low Jerk." This state of being "internally hot but externally quiet" is the physical entity of "Iki" and the state where S_{eff} of this model is maximized.

4.2 Correlation between EEG and Motion

By simultaneously measuring EEG and motion capture, we verify that the decrease in DMN activity ($\sigma \downarrow$) and the decrease in motion trajectory Jerk ($D \uparrow$) occur with significant temporal correlation. This will demonstrate that spiritual "Mushin" (No-Mind) is a necessary condition for physical "smoothness."

5 Discussion

5.1 Thermodynamic Reinterpretation of "Benevolence"

Traditionally, the act of aligning with others has been discussed in ethical contexts as "moral benevolence." However, this model reinterprets it as a "physical optimization process to minimize the free energy of the entire system." An artisan treats the subject (material or tool) "gently" not solely out of compassion, but because it is the most rational strategy to obtain the best result (maximum attractor formation power) according to physical laws. The brain's reward system is thought to have evolved to reinforce this "physical rationality" as "beauty" or "pleasure."

5.2 Extension to Social Interaction

This paper focused on the signal purification process within an individual. However, the generated S_{eff} becomes a trigger for synchronization phenomena in groups. We hypothesize that the coupling strength K in the Kuramoto model of nonlinear physics is proportional to the product of S_{eff} between individuals. Future work will discuss the mechanism by which the presence of individuals with high "Iki" promotes synchronization (Itza-Konryu) of the entire group.

6 Conclusion

This study has shown that "Iki" is not a vague poetic concept but "the most physically efficient control signal state motivated by prosocial homeostasis, generated by the elimination of neural entropy and minimum jerk control." This definition bridges Japanese traditional embodied knowledge with modern control engineering and neuroscience, providing a quantitative answer to the question of how human "Will" determines physical reality.

References

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